

LOW-LIGHTGAN: LOW-LIGHT ENHANCEMENT VIA ADVANCED GENERATIVE ADVERSARIAL NETWORK WITH TASK-DRIVEN TRAINING

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ABSTRACT

We propose a low-light enhancement method using an advanced generative adversarial network (GAN) and a task-driven training set. Unlike traditional training sets that only synthesize global illumination, we apply local illumination to make the training images. Furthermore, we enhance traditional GANs with spectral normalization and advanced loss functions, making training stable and leading to accurate results. Experimental results show that our method outperforms state-of-the-art methods qualitatively and quantitatively and alleviates saturation problems in bright areas, which typically occur after traditional low-light enhancements.

Index Terms— Low-light enhancement, task-driven training set, GANs, spectral normalization

1. INTRODUCTION

Great attention has been paid to low-level vision problems, dealing with low-light, hazy, and rainy images captured in outdoor environments. These images are known to significantly reduce image quality and deteriorate conventional computer vision algorithms. Moreover, low-light is a key reason for poor algorithmic performance, because most image contents are not visible. Thus, to utilize vision algorithms at night for autonomous driving, video surveillance, etc., low-light enhancement is necessary as a pre-processing step. If low-light enhancement is not carried out in advance, many computer vision algorithms will fail during nighttime.

Recently, owing to the evolution of deep learning, performance of computer vision algorithms has considerably enhanced. Thus, in this paper, we propose an efficient method for low-light enhancement based on generative adversarial networks (GAN), which have been extensively studied in deep-learning fields. Additionally, we present a new training set to solve saturation problems in bright areas, which easily occur with conventional low-light enhancement methods. Fig.1 shows the low-light enhancement results and demonstrates the effectiveness of the proposed task-driven training set. To construct the task-driven training images, we decompose an original image into multiple regions based on quad-tree decomposition and apply different illuminations to



Fig. 1. Low-light enhancement results: (a) Low-light images. (b) Low-lightGAN with global illumination. Synthesizing global illumination for training images can cause saturation problems in bright areas. (c) Low-lightGAN with local illumination. The proposed task-driven training set makes no saturation problems.

each decomposed region. Then, a training image is made by randomly cropping the locally illuminated image.

Our contributions can be summarized as follows:

- We propose a low-light enhancement method called low-lightGAN. To the best of our knowledge, the method is the first to utilize GAN for low-light enhancement.
- We enhance existing GAN architectures with spectral normalization techniques and advanced loss functions, facilitating stable training and producing accurate results.
- We present a new strategy of generating training images for low-light enhancement based on local illumination synthesis. This prevents low-light enhancement methods from saturating bright areas at training level.

Fig.2 shows the overall procedure of Low-lightGAN.

2. RELATED WORK

Over the last decade, histogram equalization-based methods [1] have been widely studied. To overcome their limitations, retinex theory [2, 3, 4] and high dynamic range (HDR)-based methods [5, 6] have been introduced. Retinex-based algorithms divide an image into illumination and reflectance components. For example, single-scale retinex (SSR) [7] is the representative method based on the retinex theory. Multi-scale retinex (MSR) [8] expanded SSR by incorporating a

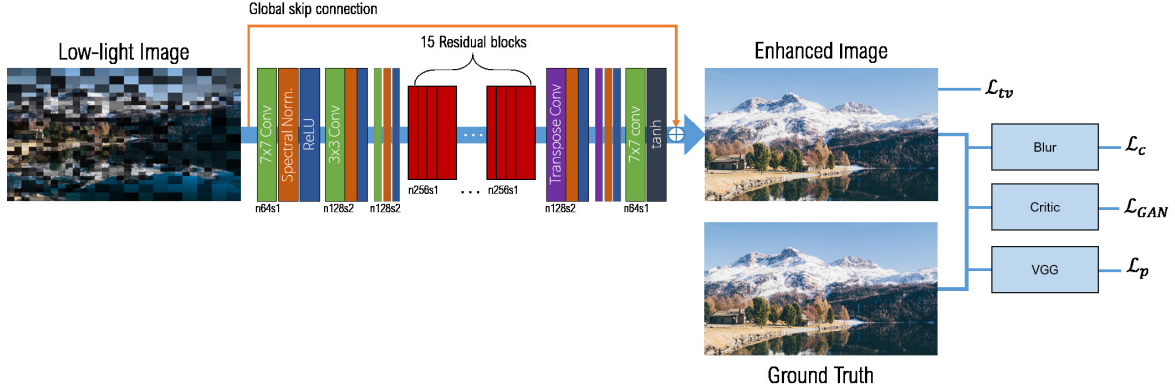


Fig. 2. Overall procedure of Low-lightGAN. Please refer to Section 3.2 for the detailed architecture and Section 3.1 for explanation of loss functions.

color restoration method. Guo *et al.* proposed a low-light enhancement method called LIME [9], which estimated the illumination using a structure prior. Recently, deep learning-based methods (*e.g.*, MSR-Net [10], LightenNet [11], Cai *et al.* [12], DeblurGAN [13], and RetinexNet [14]) have emerged. DeepFuse [15] used unsupervised learning with HDR. Particularly, the GAN proposed by Goodfellow *et al.* [16] has been the most actively studied in recent years. Arjovsky *et al.* [17] approximated Jensen-Shannon divergence through a Wasserstein distance to stably learn GANs. Gulrajani *et al.* [18] used a gradient penalty term to force the WGAN's Lipschitz condition. Miyato *et al.* [19] proposed spectral normalization to make discriminators stable during training.

In contrast to the aforementioned methods, our method efficiently solves low-light enhancement problems using adversarial learning, and successfully deals with local illumination with a new training set.

3. LOW-LIGHTGAN

For low-light enhancement, we adopt the deep neural architecture used by DeblurGAN [13]. Given an original image (*i.e.*, ground-truth), we synthesize a locally illuminated image comprising the task-driven training set. We train the proposed Low-lightGAN so that the output images (*i.e.*, low-light enhanced images) become more similar to ground-truth as training proceeds. For stable training, we apply spectral normalization to the generator of Low-lightGAN, such like self-attention GAN [20]. Additionally, we employ a Wasserstein distance with a gradient penalty for stable training. To make Low-lightGAN more accurate, we propose four loss functions, explained in the next section.

3.1. Loss Function

Using four loss functions, Low-lightGAN considerably improves the accuracy of low-light enhancement. The proposed

final loss function is

$$\mathcal{L}_{total} = \mathcal{L}_{GAN} + \lambda_1 \cdot \mathcal{L}_p + \lambda_2 \cdot \mathcal{L}_c + \lambda_3 \cdot \mathcal{L}_{tv}, \quad (1)$$

where $\mathcal{L}_{GAN}$, $\mathcal{L}_p$, $\mathcal{L}_c$, and $\mathcal{L}_{tv}$ denote the adversarial, perceptual, color, and total variation losses, respectively, and λ_1 , λ_2 , and λ_3 are weighting parameters.

Adversarial loss ($\mathcal{L}_{GAN}$): We use WGAN-GP [18], which demonstrates its effectiveness for conditional GAN, as a critic function. $\mathcal{L}_{GAN}$ is defined as follows.

$$\mathcal{L}_{GAN} = \sum_{n=1}^N -D(G(I_L^{(n)})), \quad (2)$$

where $I_L^{(n)}$ is the n -th low-light image, and $G(\cdot)$ and $D(\cdot)$ are generator and discriminator, respectively.

Perceptual loss ($\mathcal{L}_p$): This loss aims to measure the similarity between the low-light enhanced image $G(I_L)$ and ground-truth I_{GT} at the feature level, which is widely used for image quality enhancement tasks.

$$\mathcal{L}_p = \frac{1}{CHW} \|\phi(I_{GT}) - \phi(G(I_L))\|^2, \quad (3)$$

where ϕ is 14-th feature map obtained by the pre-trained VGG-19 network [21], and C , H , and W denote the channel, height, and width of 14-th feature map, respectively.

Color loss ($\mathcal{L}_c$): The color of the low-light enhanced image should be similar to ground-truth. The color loss [22] causes them to be similar, and alleviates color distortion problems.

$$\mathcal{L}_c = \|\mathcal{G}(I_{GT}) - \mathcal{G}(G(I_L))\|^2, \quad (4)$$

where $\mathcal{G}$ is a Gaussian blur function. $\mathcal{G}$ removes local details of images but preserves global information, such as color, which enables color comparison.

Total variation loss ($\mathcal{L}_{tv}$): The total variation loss aims to make the enhanced image spatially smooth.

$$\mathcal{L}_{tv} = \frac{1}{CHW} \|\nabla_x G(I_L) + \nabla_y G(I_L)\|^2, \quad (5)$$

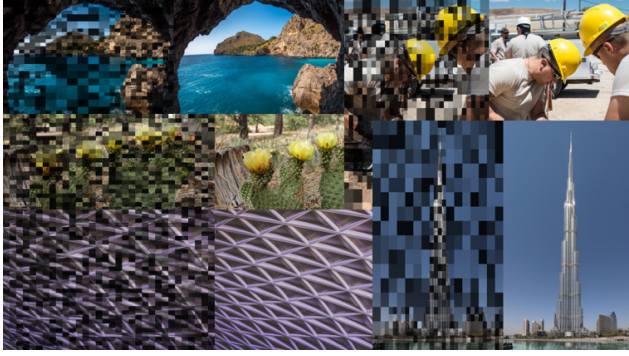


Fig. 3. Task-driven training set.

where ∇_x and ∇_y differentiate the images via the x and y directions. C , H , and W are the channel, height, and width of the enhanced image, respectively.

3.2. Network Architecture

We adopt generator and discriminator architectures of the baseline method [13], but we also use 15 residual blocks to extract more accurate feature maps. Furthermore, the instance normalization [23] of generator and discriminator is replaced by spectral normalization [19]. The global-skip connection is also applied to the network by adding the original image to the low-light enhanced image obtained by the generator. For more details, please refer to [13].

3.3. Task-driven Training Set

The main bottleneck of low-level vision, based on deep learning, is the absence of useful training sets. To alleviate this problem, traditional methods synthesize training images for each task (*e.g.*, fog, low light) by using clean images. For example, MSR-Net [10] and LightenNet [11] synthesize low-light globally. However, real-world images typically contain global and local illumination at the same time. If only global illumination is applied to the image, the overall brightness is improved, but saturation problems can occur near bright areas. Our method solves this problem by synthesizing local illumination at the training-set level. For this, we use a quad-tree decomposition technique to divide the image into multiple regions, in which each region is typically homogeneous. Then, we synthesize different illuminations to each region. To make training images, we randomly crop the locally illuminated image. Thus, each training image can have locally different degrees of illumination. The generated training images are shown in Fig.3.

The quad-tree decomposition is terminated when the min-max difference of pixels in the patch is smaller than a predefined value. The degree of illumination is divided into 10 levels, ranging from 0.1 to 1, where 1 indicates the original brightness without any illumination synthesis. By making



Fig. 4. Advantages of using global skip connection: The original image (first row) is added to the result from generator (second row) via global-skip connection. The generated image at the second row has very low values at the originally bright regions. Thus, the addition of two images by global-skip connection prevents the originally bright areas from saturating after low-light enhancement.

some regions to preserve the original brightness during illumination synthesis, our method prevents saturation problems in originally bright areas after low-light enhancement. We utilize the DIV2K dataset [24] to synthesize illumination. This is originally used for super resolution. This dataset contains high quality images of people, objects, and backgrounds.

4. EXPERIMENTS

Our Low-lightGAN was compared to recent state-of-the-art methods, namely, RetinexNet [14] and LIME [9]. For evaluation, the naturalness image quality evaluator (NIQE) [25] was used, which is a no-reference metric that can measure image quality without ground-truth. During training, we randomly cropped the 100×100 patches using 900 images with 2K resolution. We used a batch size of 15 and the ADAM optimizer. We trained the network for 300 epochs on an NVIDIA GeForce GTX TITAN Xp GPU. A learning rate was set to $1e - 5$. The parameters λ_1 , λ_2 , and λ_3 in (1) were set to 0.1, 25, and 400, respectively.

4.1. Ablation Study

Fig.1 shows the results when global illumination was applied and when the proposed local illumination was applied to the training images. Using the training set with global illumination, the sky areas were saturated after low-light enhancement, whereas Using the training set with the local illumination preserved original brightness and details of the sky. Fig.4 demonstrates the advantage of using the global-skip connection.

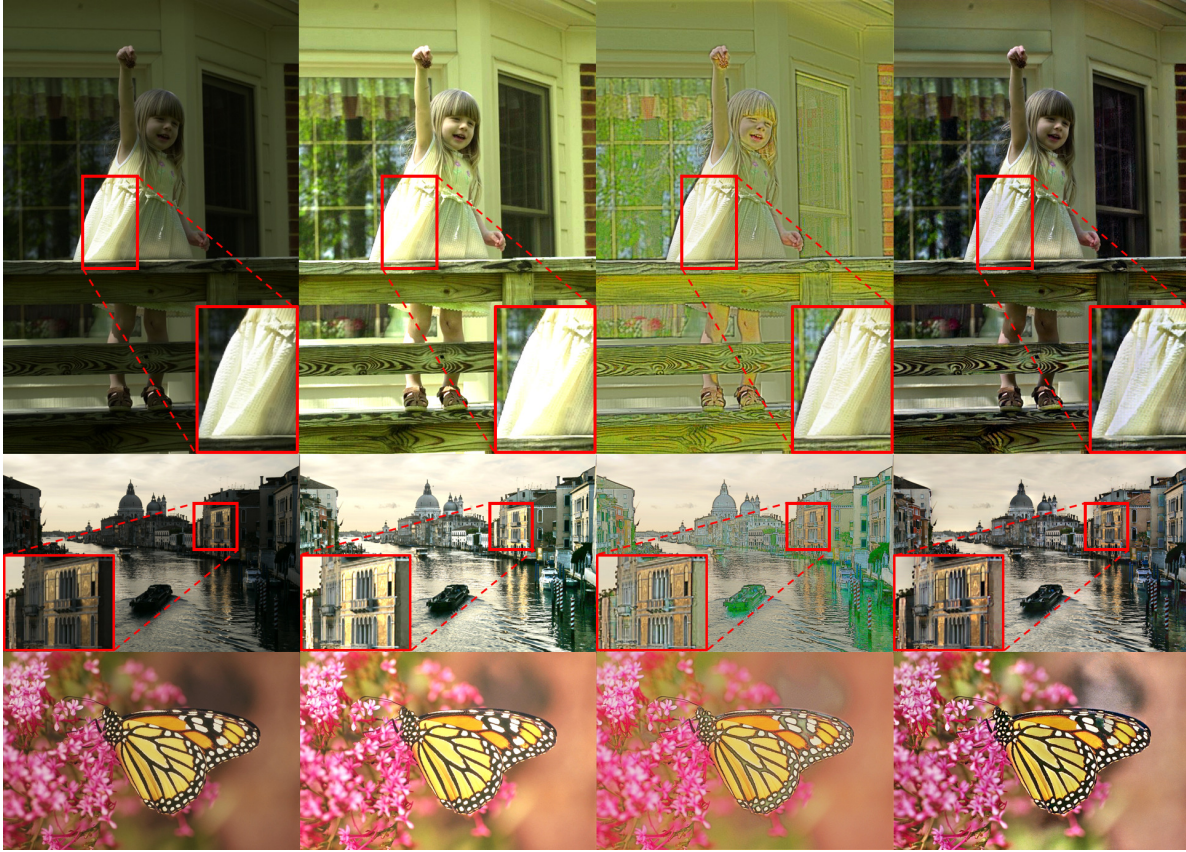


Fig. 5. Comparison to other methods: Left to right: input image, LIME [9], RetinexNet [14], and Low-lightGAN.

4.2. Comparison with Other methods

Fig.5 shows qualitative results of Low-lightGAN and other algorithms. Real-world images typically contain normally illuminated and weakly illuminated areas, concurrently. In the first row of Fig.5, the area around the clothes is bright. For LIME, the overall brightness was improved naturally, but saturation problems occurred in original bright areas. For Low-lightGAN, global brightness was improved less than LIME, but the method successfully recovered local brightness and acquired a natural image with a similar tone as the original. Moreover, the saturation did not occur in clothing parts. RetinexNet produced unnatural results, as shown in the first and second rows of Fig.5. In the last row, LIME enhanced overall brightness but lost local details. On the other hand, Low-lightGAN produced clearer images while maintaining the local details. However, the generator of Low-lightGAN attempted to generate some objects in originally dark areas that have a value of zero. Thus, the artifacts were also generated in the background. This is a chronic problem of generative methods.

Table 1 shows a quantitative comparison in terms of NIQE. Low-lightGAN quantitatively outperforms other methods, which demonstrates that the proposed four-loss functions and task-driven dataset are useful for low-light enhancement.

Table 1. NIQE scores [25]. The smaller the value of NIQE is the more natural result.

	LIME [9]	RetinexNet [14]	Low-lightGAN
NASA 15	2.298	3.516	2.467
grandcanal	2.306	2.795	2.856
monarch	2.894	2.970	2.649

5. CONCLUSION

In this paper, we proposed a low-light enhancement method based on advanced GAN and a task-driven training set. Unlike traditional training dataset synthesis using global illumination, the proposed training set with local illumination solved saturation problems in bright areas. We enhanced existing GANs using spectral normalization and advanced loss functions. Experimental results showed that Low-lightGAN achieved the state-of-the-art performance qualitatively and quantitatively.

6. ACKNOWLEDGMENT

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